Practical No : 14

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Q.14) Write a C program for DFS.

```
#include <stdio.h>
#define MAX 20
int graph[MAX][MAX];
int visited[MAX];
int n;
void DFS(int vertex)
{
   int i;
   // Mark the current vertex as visited
   visited[vertex] = 1;
   printf("%d ", vertex + 1);
   // Visit all adjacent vertices
   for (i = 0; i < n; i++) {
      if (graph[vertex][i] == 1 && visited[i] == 0) {
         DFS(i);
      }
   }
}
int main()
{
   int i, j, start;
   printf("Enter number of vertices: ");
   scanf("%d", &n);
   printf("Enter the adjacency matrix:\n");
   for (i = 0; i < n; i++) {
      for (j = 0; j < n; j++) {
```

```
            scanf("%d", &graph[i][j]);
        }
    }
    // Initialize all vertices as unvisited
    for (i = 0; i < n; i++) {
        visited[i] = 0;
    }
    printf("Enter starting vertex: ");
    scanf("%d", &start);
    printf("\nDFS Traversal: ");
    DFS(start - 1);
    printf("\n");
    return 0;
}
```

Output

